Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State what is meant by a member in tension.

- The member is being pulled / stretched by the forces at its ends

3 Quick check

- a) 980 N
- b) 1.0 m
- c) CWM = ACWM
- d) A triangle can't change shape under load

4 Key terms

- 1 – B
- 2 – F
- 3 – A
- 4 – D
- 5 – C
- 6 – E

5 State the two conditions of equilibrium.

- The sum of the forces = 0 (upward = downward, left = right)
- The sum of the moments = 0 (clockwise = anticlockwise)

6 Calculate the reactions at each support.

- $F = mg = 95 \times 9.8 = 931$ N
- Moments about the left support: $931 \times 9 = R_{right} \times 25$
- $R_{right} = 335$ N
- $R_{left} = 931 - 335 = 596$ N